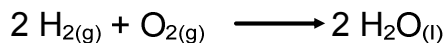


Limiting & Excess Reagents

In the synthesis of water, we require *exactly* 2 mol of H₂ and *exactly* 1 mol O₂ to create *exactly* 2 mol H₂O



What would happen if we have 2 mol of H₂ and **8 mol** of O₂?

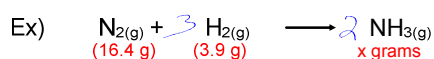
Would this change the amount of water that would be created?

NO! H₂ still runs out first.

A chemical reaction will stop once any one of the reactants runs out. This reactant is known as the limiting reagent. Any other reactant is an excess reagent.

Steps to stoichiometry with a limiting reagent:

- 1) Balance the chemical equation.
- 2) Assume one of the reagents as excess and one as limiting.
- 3) Calculate how much of the excess reagent is consumed if all of the limiting reagent is used. This will determine if the assumption was correct.
- 4) Complete stoichiometric calculation using the limiting reagent.



Assume H_2 is limiting

$$m_{\text{N}_2} = 3.9 \text{ g H}_2 \times \frac{1 \text{ mol H}_2}{2.02 \text{ g}} \times \frac{1 \text{ mol N}_2}{3 \text{ mol H}_2} \times \frac{28.02 \text{ g}}{1 \text{ mol N}_2}$$

$$= 18 \text{ g}$$

$\therefore \text{N}_2$ is limiting

Assume N_2 is limiting

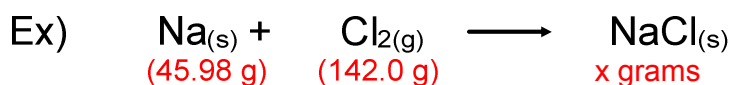
$$m_{\text{H}_2} = 16.4 \text{ g N}_2 \times \frac{1 \text{ mol N}_2}{28.02 \text{ g}} \times \frac{3 \text{ mol H}_2}{1 \text{ mol N}_2} \times \frac{2.02 \text{ g}}{1 \text{ mol H}_2}$$

$$= 3.55 \text{ g}$$

$\therefore \text{N}_2$ is limiting

$$m_{\text{NH}_3} = 16.4 \text{ g N}_2 \times \frac{1 \text{ mol N}_2}{28.02 \text{ g}} \times \frac{2 \text{ mol NH}_3}{1 \text{ mol N}_2} \times \frac{17.04 \text{ g}}{1 \text{ mol NH}_3}$$

$$= 19.9 \text{ g}$$



$$m_{\text{Cl}_2} = \frac{45.98 \text{ g}}{1} \times \frac{1 \text{ mol Na}}{22.98 \text{ g}} \times \frac{1 \text{ Cl}}{2 \text{ Na}} \times \frac{70.86}{1 \text{ mol Cl}}$$

$$= 70.89 \text{ g} \therefore \text{Na is limiting}$$

$$m_{\text{NaCl}} = \frac{45.98 \text{ g Na}}{1} \times \frac{1 \text{ mol Na}}{22.98 \text{ g}} \times \frac{1 \text{ mol NaCl}}{1 \text{ mol Na}} \times \frac{58.44}{1 \text{ mol NaCl}}$$

$$= 116.9 \text{ g of NaCl}$$

Ex) 15.00 g of aluminum sulfide added to 10.00 g of water.

- a) How much aluminum hydroxide is produced?
- b) How much excess reagent is left?

